The auxiliary torus of 7+1 spacetime: provenance

The task, the decisions, the inputs, the data, the derivations and the verification record of the document on the auxiliary torus

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states what was asked, how it was interpreted, every decision taken (among them the new section 25 of 7+1.nb, which computes every worked example), which results of the earlier sections and which data were used and where they come from, how each result was derived, and how it was checked: the 27 checks of section 25 (27 True), the 38 boolean checks of the independent Python program tools/ref_torus_brane.py (38 True), the comparison of the 17 groups of numbers printed by both programs, the check that the earlier sections of the notebook are unchanged, the program versions and the SHA-256 hashes of the files.

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1. The task

On 28 October 2026 the author asked for the following (summarized; the original wording is the author's private working material and is not reproduced):

1. Define and explain the "auxiliary torus" precisely and correctly, in its own documentation file with its own symbol file and provenance file, each in Markdown and LaTeX with the LaTeX compiled to PDF.
2. Explain it for students; be clear, use examples, and organize the material well.
3. Do the same, in a separate set of three files, for the "world brane".
4. Push these files, and any other relevant files, to the repository github.com/once-ere/7plus1.
5. Then, in a second request: read a note of the author on the radion (a private PDF in the workspace, not part of the repository), use its ideas to verify and clarify the radion and brane documents in preparation, and create examples in which the four extra space dimensions are (1) a 4-sphere and (2) a 4-torus. For this document that means the comparison of the torus with a 4-sphere of the same volume (section 10 of the document) and the precise definition of the radion used in its sections 6.4–6.5 and 10.1.

The standing rules of the project apply: every documentation file has a companion provenance file and a symbol file, in Markdown and LaTeX, compiled to PDF; no file refers the reader to another documentation, provenance or symbol file; every equation that is created or used is written in the file that uses it. The repository's .gitignore is respected.

Interpretation. The auxiliary torus is the compact space of the four auxiliary coordinates $x_4, \ldots, x_7$ of 7+1.nb, on which sections 17–24 rely (the Kaluza–Klein masses, the volume $V_\perp$, the radion, the Casimir energy). "Precisely" is read as: a definition as a quotient space, the metric quantities (size, volume) with their equations, its symmetries, and the consequences that the model uses, each stated as an equation; "correctly" as: every worked example is computed and checked twice, by the notebook and by an independent program. The coordinates of 7+1.nb are kept: $X = (x_0, \ldots, x_7)$, x_0 the time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four auxiliary space-like coordinates.

1.1 The signature: the author's explanation of 27 October 2026

Source. On 27 October 2026 the author of the project explained that the xAct list $\{7, 1, 0\}$ declares seven space-like dimensions and one time-like dimension, while $\mathrm{diag}(+1, -1, \ldots, -1)$ is simply the squares of the octonion basis, and asked that this be made clear in all documentation, examples and provenance records. **What it corrected.** The version of 7+1.nb pushed in the commits 52d8729 to dc2f178 of github.com/once-ere/7plus1 declared the xAct metric of section 12 with `DefMetric[{1, 7, 0}, …]`, reading the list as the signs of η; since then section 12 declares `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`, exactly as the author's notebook Einstein-Lovelock-7+1-Nash.nb does. **What was added.** Checks in 7+1.nb — section 4.7: η is the squares of the octonion basis, $e_A e_A = \eta_{AA} e_0$ and $\mathrm{Re}(e_A e_B) = \eta_{AB}$; section 12.2: `SignatureOfMetric[g71]` $= \{7, 1, 0\}$ with `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$, and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$; section 17: the flat metric $\mathrm{diag}(+1, -1, -1, -1)$ of the world brane is the squares of the basis e_0, e_1, e_2, e_3 of the quaternion subalgebra — and the same statements in the Python programs `tools/ref71.py`, `tools/ref_gravity.py` and `tools/ref_sm.py`; the explanation itself in the text of 7+1.nb (sections 1.2, 1.4, 4.2, 4.7, 7, 11, 12.2, 13, 15.1, 17) and in every document, symbol file and provenance file of the repository. **What did not change.** No result. Compared output by output with the backup below, every output of 7+1.nb is the same, except the declared list itself, the new checks and the check counts (and, as in every run, the name of the private context and the measured times); in particular every component of every Lovelock tensor is unchanged. The components follow from the component matrix of the metric, and the sign of the determinant, `SignDetOfMetric`, is -1 for both lists. **Backup.** The evaluated notebook as it was before is kept in the workspace as `backups/7+1_2026-10-27_before_signature.nb` (the folder backups/ is excluded from the repository by .gitignore), 1034401 bytes, SHA-256 `060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`; `tools/verify_7p1_nb.py` compares every input, text and section cell of 7+1.nb with it and finds every cell present, each one unchanged or changed by this explanation, and it compares every output in the same way.

The explanation, in full. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta = \det \eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2. Decisions

#	decision	reason / equation
1	append section 25 to 7+1.nb ("The auxiliary torus and the world brane: definitions and worked examples"); the summary of all checks becomes section 26	every worked example of the document is then computed and checked by the notebook itself, in the same evaluation as everything else
2	write an independent Python program, tools/ref_torus_brane.py, for the same examples	its own lattice sieve, Fourier transform (mpmath), symbolic integrals and curvature (sympy); no code shared with the notebook
3	back up the evaluated notebook before appending section 25 and check that sections 0–24 are unchanged	tools/verify_7p1_nb.py compares every cell and output with the backup (section 6.3)
4	the torus of 7+1.nb is the cubic torus $\mathbb{R}^4/(L_c\mathbb{Z})^4$ with periodic fermions	the choice of sections 18 and 22–24; other lattices and spin structures are named but not used
5	keep both L_c and b, and state that only bL_c is physical	7+1.nb uses both symbols; $\vec{y} = \lambda\vec{y}\,'$ maps $(L_c, b) \to (L_c/\lambda, \lambda b)$
6	the size is defined by $b^8 = \det(-g_{kl})$	the definition of 7+1.nb (section 20.1); for the metrics of the model $-g_{kl} = b^2\delta_{kl}$
7	the Newton-law example in linearized gravity without the stabilizing potential; the stabilizer is added through the radion mass	the simplest setting in which images and Kaluza–Klein modes can be compared exactly
8	the image sum is summed over the shells $k \le k_{\max}$ of section 22.2 and closed by a continuum tail; the Kaluza–Klein sum over the shells $k \le 20000$	tail and truncation errors are far below the tolerance 10^{-8} (section 4)
9	the Newtonian potential is written $\Phi_{\rm N}$ (not Φ), the PDG radius $R_{\rm PDG}$ (not R)	Φ is the Higgs doublet and R the scalar curvature in 7+1.nb
10	numbers today with $M_8 = 10$ TeV, $M_{\rm Pl} = 2.435 \times 10^{18}$ GeV, $m_r = 10$ GeV	the choices of sections 22–24
11	the check label "7.7 10^17 states below m_h" of section 24.4 is left unchanged; the document states its exact meaning	it is the weighted sum $\sum(1 - \mu_{\vec{n}}^2/m_h^2)^2 = V_\perp m_h^4/(192\pi^2)$, not the number $N(m_h) = V_\perp m_h^4/(32\pi^2)$ of levels; changing a label of section 24 would change a verified section
12	the orbits of the shells are computed with $\lvert n_i\rvert \le 3$	enough for $k \le 12$, since $\lvert n_i\rvert^2 \le k$
13	compare the torus with a 4-sphere of the SAME volume $V_\perp$ (section 26 of 7+1.nb, appended for the second request)	the same four-dimensional Newton constant $8\pi G_4 = \kappa/V_\perp$; the differences are then those of the geometry alone
14	the radion in the Einstein frame, $ds^2 = e^{-4\beta}g_E - e^{2\beta}\gamma$, on both spaces	the conventions of section 20.3 of 7+1.nb; the reduction is checked in section 26.1

3. Inputs from the earlier sections of 7+1.nb

object	Mathematica	section	equation or value
cosmological metric	`gCos`, `a3[x0]`, `b4[x0]`, `cvCos`	11.2	$ds^2 = dx_0^2 - a^2 d\vec{x}^2 - b^2 d\vec{y}^2$; $R = -2(3H_a' + 6H_a^2 + 4H_b' + 10H_b^2 + 12H_aH_b)$
inhomogeneous metric	`gPoly`, `cvPoly`	11.3	$n = 1 + \frac{x_0x_4}{5} + \frac{x_0^2}{11}$, $a = 1 + \frac{x_0}{3} + \frac{x_4^2}{7}$, $b = 2 - \frac{x_0x_4}{4} + \frac{x_0^2}{9} + \frac{x_4}{13}$
Dirac matrices, brane representation, intertwiner	`Γ17`, `g4D`, `g5D`, `rhoK`, `rBr`, `uInt`	3–5, 17.2	$\Gamma_A U = U R_A$, $R_\mu = \gamma_\mu \otimes 1_4$, $R_k = \gamma_5 \otimes \rho_k$
Kaluza–Klein operator	`xKK`	18.1	$X = \gamma_5 \otimes (\vec{\rho}\cdot\vec{q})/b_0$, $X^2 = -\lvert\vec{q}\rvert^2/b_0^2$
stress tensor from a comoving energy	`feStress[W]`	20.2	$\varepsilon = W/(a^3b^4)$, $p_a = -\partial_a W/(3a^2b^4)$, $p_b = -\partial_b W/(4a^3b^3)$
radion, stabilizing potential, four-dimensional constants	`uPot`, `kU`	20.3	$U = U_0 + \frac{K_U}{2}\beta^2$, $m_r^2 = \kappa K_U/12$, $8\pi G_4 = \kappa/V_\perp$, $\phi = \sqrt{12}M_{\rm Pl}\beta$
Kasner exponents	`rIp1`, `rIq1`, `rIp2`, `rIq2`	21.1	$p_1 = (1+2\sqrt{2})/7$, $q_1 = (2-3\sqrt{2})/14$, $3p_2 + 4q_2 = 3$
cubic regime	`rIcub`, `rIhStar`	21.2–21.3	$h_0 = -\sqrt{\alpha_2/(30\alpha_3)}$, $h_* = -\sqrt{\alpha_2/(54\alpha_3)}$
decompactification without stabilizer	`rIlate`	21.3	$\tau H_b \to \frac{2}{5}$, $p_b/\varepsilon \to \frac{1}{4}$
capture of the radion	`rIIaF`, `rIIphF`, `rIItEF`	22.1	$\beta \simeq A\sin(m_r\tau + \vartheta)/(m_r(\tau - \tau_E))$
Jacobi's $r_4(k)$, the volume today	`rIIr4`, `rIIsig`, `rIIkmax`, `rIIVp`	22.2	$r_4 = 8\sigma(k) - 32\sigma(k/4)$; $V_\perp = M_{\rm Pl}^2/M_8^6$
the radion today	—	22.4	$\beta_{\rm eq}$ from Λ_4 and matter, from a tension $f_W = M_8^4$

object	Mathematica	section	equation or value
the far future	`rIIIH`	23.1–23.2	$H_\infty = H_0\Omega_\Lambda^{1/2}$; eight-dimensional de Sitter $21H^2 - 420\alpha_2 H^4 + 2520\alpha_3 H^6 = \Lambda$
tower sums, Casimir energy, ρ_Λ	`dkTowN`, `dkCas`, `dkRhoL`	24.4–24.5	$\sum(1-\mu^2/m_h^2)^2 = V_\perp m_h^4/(192\pi^2)$; $\rho_{\rm Cas}$; $3H_0^2 M_{\rm Pl}^2\Omega_\Lambda$
the radion on T^4 and S^4, the sphere today	`rdV`, `rdDl`, `rdZon`, `rdKS`, `rdRS4`, `rdToday`	26.1–26.4	$V(\beta) = M_{\rm Pl}^2[e^{-4\beta}(\Lambda + \kappa U) - 6\sigma e^{-6\beta}/r_0^2]$; $d_l = (2l+3)(l+1)(l+2)/6$; $r_{S^4} = (3V_\perp/(8\pi^2))^{1/4}$

Before section 25 was appended, the evaluated notebook (sections 0–24 and the summary of all checks, then numbered 25) was copied to `backups/7+1_2026-10-28_before_torus_brane.nb` (1940664 bytes, SHA-256 `39f64ee81d35c0c9fb94188fd7f8c8e75c96ea198fbf72a5ca6b3c0027426018`; excluded from the repository by .gitignore).

4. Data and their sources

quantity	value used	source
$\hbar c$	$1.973269804 \times 10^{-16}$ GeV m	CODATA 2018 (E. Tiesinga et al., Rev. Mod. Phys. 93 (2021) 025010)
proton charge radius r_p	0.8414 fm	CODATA 2018
Bohr radius a_0	$5.29177210903 \times 10^{-11}$ m	CODATA 2018
$M_{\rm Pl}$ (reduced)	2.435×10^{18} GeV	$(8\pi G_N)^{-1/2}$, as in sections 17–24
M_8, m_r	10 TeV, 10 GeV	the choices of sections 22–24 (natural values)
m_h	125.20 GeV	Particle Data Group 2025 (as in section 17.3)
bounds on M_D	5.9 to 11.2 TeV for 6 to 2 extra dimensions	Particle Data Group, Phys. Rev. D 110 (2024) 030001 (ATLAS, 139 fb^{-1})
ρ_Λ	2.518×10^{-47} GeV^4	$3H_0^2 M_{\rm Pl}^2\Omega_\Lambda$ with the Planck 2018 values of section 22.4
α_3/α_2^2	10^{-6}	the choice of sections 21–22.1

Software: Wolfram Language 15.0.1 (front end WolframNB.exe; the kernel configuration "Eval71" is a fresh kernel); Python 3.13.17 with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, mpmath 1.3.0.

5. Derivations

The volume. For $-g_{kl} = b^2\delta_{kl}$, $\det(-g_{kl}) = b^8$ and $V_\perp = \int_{[0,L_c)^4}\sqrt{\det(-g_{kl})}\,d^4y = b^4L_c^4$; under $\vec{y} = \lambda\vec{y}\,'$ the line element $b^2 d\vec{y}\cdot d\vec{y} = (\lambda b)^2 d\vec{y}\,'\cdot d\vec{y}\,'$, so $(L_c, b) \to (L_c/\lambda, \lambda b)$ and bL_c, $b^4L_c^4$ are unchanged.

The orbits. A signed permutation permutes the entries of $\vec{n}$ and changes their signs; the orbit of $\vec{n}$ is the set of vectors with the same multiset $\{|n_i|\}$. Its size is the number of distinct arrangements of the multiset times 2^z, z the number of non-zero entries:

$$|\mathrm{orbit}(\vec{n})| = \frac{4!}{\prod_j m_j!}\,2^z, \qquad \text{e.g. } (1,1,0,0):\ \frac{4!}{2!\,2!}\cdot 2^2 = 24, \quad (1,1,1,1):\ 1\cdot 2^4 = 16.$$

Orthonormality and Parseval. $\int_0^{L_c} e^{2\pi i(m-n)y/L_c}dy = L_c\delta_{mn}$, so $\int_F b^4\phi_{\vec{n}}^*\phi_{\vec{m}} = b^4L_c^4\delta_{\vec{n}\vec{m}}/V_\perp = \delta_{\vec{n}\vec{m}}$. With $\cos^2\theta = \frac{1}{2} + \frac{1}{4}e^{2i\theta} + \frac{1}{4}e^{-2i\theta}$ the bump $f = \prod\cos^2(\pi y_k/L_c)$ has $c_{\vec{n}} = \sqrt{V_\perp}\prod w(n_k)$, and

$$\sum_{\vec{n}}|c_{\vec{n}}|^2 = V_\perp\Big(\frac{1}{4}+\frac{1}{16}+\frac{1}{16}\Big)^4 = \Big(\frac{3}{8}\Big)^4 V_\perp, \qquad \int_F b^4 f^2\,d^4y = b^4\Big(\int_0^{L_c}\cos^4\frac{\pi y}{L_c}dy\Big)^4 = b^4\Big(\frac{3}{8}L_c\Big)^4.$$

Kaluza–Klein masses. For $g = \mathrm{diag}(1,-1,-1,-1,-b_0^2,\ldots,-b_0^2)$, $\Box = \partial_0^2 - \nabla_{\vec{x}}^2 - b_0^{-2}\nabla_{\vec{y}}^2$; on $e^{-iEx_0 + i\vec{k}\cdot\vec{x} + i\vec{q}\cdot\vec{y}}$ this is $-E^2 + |\vec{k}|^2 + |\vec{q}|^2/b_0^2$, so $E^2 = |\vec{k}|^2 + m^2 + |\vec{q}|^2/b_0^2$ and $\mu_{\vec{n}} = |\vec{q}_{\vec{n}}|/b_0 = 2\pi|\vec{n}|/(b_0L_c) = 2\pi|\vec{n}|/\ell_\perp$. For dirac16complex, $(\gamma\cdot p\otimes 1_4 + X)^2 = (\gamma\cdot p)^2\otimes 1_4 + X^2$ because γ_5 anticommutes with every γ_μ, and $X^2 = \gamma_5^2\otimes(\vec{\rho}\cdot\vec{q})^2/b_0^2 = -|\vec{q}|^2/b_0^2$ from $\rho_k\rho_l + \rho_l\rho_k = -2\delta_{kl}$.

The graviton states. A massless particle in D dimensions has the little group $SO(D-2)$; the graviton is its symmetric traceless tensor, $\frac{(D-2)(D-1)}{2}-1=\frac{D(D-3)}{2}=20$ states. At the zero mode the $SO(2)$ of four dimensions sees $g_{\mu\nu}$ (2), four vectors $g_{\mu k}$ (4×2) and the symmetric g_{kl} (10). At a massive level with $\vec{q}\neq 0$ the little group of the four-dimensional massive particle is $SO(3)$, and the 20 states form a massive spin 2 (5), three massive vectors (the torus directions orthogonal to $\vec{q}$; 3×3) and the 6 components of a symmetric 3×3 tensor.

Jacobi and the lattice sums. $\sum_{k\geq 1}\sigma(k)k^{-s}=\zeta(s)\zeta(s-1)$ and $\sum_k \sigma(k/4)k^{-s}=4^{-s}\zeta(s)\zeta(s-1)$, so

$$\sum_{k\geq 1}\frac{r_4(k)}{k^s}=8(1-4^{1-s})\,\zeta(s)\,\zeta(s-1):\qquad Z(\tfrac{5}{2})=7\,\zeta(\tfrac{5}{2})\,\zeta(\tfrac{3}{2}),\qquad Z(4)=8(1-4^{-3})\,\zeta(4)\,\zeta(3).$$

Weyl's law and the level count. On average $r_4(k)\approx \frac{d}{dk}\big(\frac{\pi^2}{2}k^2\big)=\pi^2 k$, so $N(R)\simeq\frac{\pi^2}{2}R^4$; a level below μ has $|\vec{n}|\leq R=\mu\ell_\perp/(2\pi)$:

$$N(\mu)\simeq\frac{\pi^2}{2}\Big(\frac{\mu\ell_\perp}{2\pi}\Big)^4=\frac{V_\perp\mu^4}{32\pi^2},\qquad \sum_{\mu_{\vec{n}}<m_h}\Big(1-\frac{\mu_{\vec{n}}^2}{m_h^2}\Big)^2\simeq\frac{V_\perp m_h^4}{8\pi^2}\int_0^1 u^3(1-u^2)^2du=\frac{V_\perp m_h^4}{192\pi^2}.$$

From eight to four dimensions. For $g=\mathrm{diag}(g^{(4)}_{\mu\nu}(x),-b_0^2\delta_{kl})$ the Christoffel symbols with an index k vanish (nothing depends on $\vec{y}$ and b_0 is constant), so $R=R^{(4)}$ and $\sqrt{|g|}=b_0^4\sqrt{|g^{(4)}|}$; integrating $-\frac{1}{2\kappa}\int\sqrt{|g|}R$ over the torus gives $-\frac{V_\perp}{2\kappa}\int d^4x\sqrt{|g^{(4)}|}R^{(4)}=-\frac{1}{16\pi G_4}\int d^4x\sqrt{|g^{(4)}|}R^{(4)}$, i.e. $8\pi G_4=\kappa/V_\perp$.

Newton's law in D dimensions. The linearized Einstein equation for a static source gives $\nabla^2\Phi_{\rm N}=8\pi G_D\frac{D-3}{D-2}\rho$; with $\nabla^2 r^{-(d-2)}=-(d-2)\Omega_{d-1}\delta^d$ ($d=D-1$, $\Omega_{d-1}=2\pi^{d/2}/\Gamma(d/2)$):

$$\Phi_{\rm N}=-\frac{8\pi G_D(D-3)M}{(D-2)(D-3)\Omega_{D-2}r^{D-3}}=-\frac{\kappa M}{(D-2)\Omega_{D-2}r^{D-3}}:$$
$$D=8:\quad -\frac{\kappa M}{6\cdot\frac{16\pi^3}{15}r^5}=-\frac{5\kappa M}{32\pi^3 r^5},\qquad D=4:\quad -\frac{GM}{r}.$$

Images and Kaluza–Klein modes. The Poisson summation formula $\sum_{\vec{n}}f(\vec{n})=\sum_{\vec{m}}\hat{f}(2\pi\vec{m})$ with $\hat{f}(\vec{p})=\int d^4y\,e^{-i\vec{p}\cdot\vec{y}}f(\vec{y})$, and for a radial function in four dimensions

$$\hat{f}(p)=\frac{4\pi^2}{p}\int_0^\infty y^2J_1(py)f(y)\,dy.$$

For $f=(r^2+y^2)^{-5/2}$ the integral is the case $\nu=1$, $\mu=\frac{3}{2}$ of

$$\int_0^\infty\frac{y^{\nu+1}J_\nu(py)}{(y^2+r^2)^{\mu+1}}dy=\frac{p^\mu r^{\nu-\mu}}{2^\mu\Gamma(\mu+1)}K_{\nu-\mu}(pr)\quad(-1<\nu<2\mu+\tfrac{3}{2}),$$

with $\Gamma(\frac{5}{2})=\frac{3}{4}\sqrt{\pi}$ and $K_{-1/2}(z)=K_{1/2}(z)=\sqrt{\pi/(2z)}\,e^{-z}$:

$$\frac{p^{3/2}r^{-1/2}}{2^{3/2}\cdot\frac{3}{4}\sqrt{\pi}}\sqrt{\frac{\pi}{2pr}}\,e^{-pr}=\frac{p}{3r}e^{-pr},\qquad \hat{f}(p)=\frac{4\pi^2}{p}\cdot\frac{p}{3r}e^{-pr}=\frac{4\pi^2}{3r}e^{-pr},$$

hence $\sum_{\vec{n}}(r^2+|\vec{n}|^2)^{-5/2}=\frac{4\pi^2}{3r}\sum_{\vec{m}}e^{-2\pi r|\vec{m}|}$. **The numerical evaluation.** The left side is summed over the shells $k\leq K=k_{\max}$ and closed with the continuum tail ($r_4(k)\to\pi^2k$, $u_0=r^2+K+\frac{1}{2}$)

$$\sum_{k>K}\frac{r_4(k)}{(r^2+k)^{5/2}}\simeq\pi^2\int_{u_0}^\infty\frac{t-r^2}{t^{5/2}}dt=\pi^2\Big(\frac{2}{\sqrt{u_0}}-\frac{2r^2}{3u_0^{3/2}}\Big),\qquad \sum_{k>K}\frac{r_4(k)}{k^{5/2}}\simeq\frac{2\pi^2}{\sqrt{K+\frac{1}{2}}},$$

and the right side over $k\leq 20000$ (the omitted terms are below $e^{-2\pi r\sqrt{20000}}<10^{-96}$ for $r\geq\frac{1}{4}$). Near the source $S(r)=r^{-5}+\sum_{\vec{n}\neq 0}(r^2+|\vec{n}|^2)^{-5/2}=r^{-5}+Z(\frac{5}{2})+O(r^2)$; far from it $\frac{3r}{4\pi^2}S(r)=1+8e^{-2\pi r}+24e^{-2\sqrt{2}\pi r}+\ldots$

The factor $\frac{5}{3}$. Restoring $\ell_\perp$: $\Phi_{\rm N}\to-\frac{5\kappa M}{32\pi^3}\cdot\frac{4\pi^2}{3V_\perp r}=-\frac{5\cdot 8\pi G_4 M}{24\pi r}=-\frac{5}{3}\frac{G_4M}{r}$. In terms of exchanged particles: one-graviton exchange between two sources in D dimensions is proportional to $T_{AB}T'^{AB}-TT'/(D-2)$; for two masses at rest ($T_{00}=M$, $T=M$) this is $MM'(D-3)/(D-2)$: $\frac{5}{6}MM'$ for the zero mode of eight-dimensional

gravity, $\frac{1}{2}MM'$ for the four-dimensional graviton. The ratio is $\frac{5}{3}$; the difference, $\frac{1}{3}MM' = \frac{2}{3}\cdot\frac{1}{2}MM'$, is the radion, whose coupling $\phi T^{\mu}{}_{\mu}/(\sqrt{3}M_{\rm Pl})$ gives the potential $-\frac{MM'}{3M_{\rm Pl}^2}\frac{1}{4\pi r} = -\frac{2}{3}\frac{G_4 MM'}{r}$ ($8\pi G_4 = M_{\rm Pl}^{-2}$).

The radion coupling. With $g_E = (b/b_0)^4 h$ the four-dimensional Einstein frame (the Planck mass is $\propto b^4$), $h = e^{-4\beta}g_E$; varying $h^{\mu\nu} = e^{4\beta}g_E^{\mu\nu}$ by $4\beta g_E^{\mu\nu}$ in $\delta S = \frac{1}{2}\int\sqrt{|h|}T_{\mu\nu}\delta h^{\mu\nu}$ gives $2\beta T^{\mu}{}_{\mu}$, and $\beta = \phi/(\sqrt{12}M_{\rm Pl})$ gives $\phi T^{\mu}{}_{\mu}/(\sqrt{3}M_{\rm Pl})$ (section 20.3 of 7+1.nb).

The radion today. Static and linear in β, the 44 equation of section 20.3, $\frac{\kappa}{4}bU'(b) = \Lambda + \kappa U + 2\kappa f_W/V_\perp + \frac{\kappa}{2}(\varepsilon - 3p_a) + \kappa p_b - 12\alpha_2 H_a^2(H_a' + H_a^2)$, with $bU' = K_U\beta$ and $\frac{\kappa K_U}{4} = 3m_r^2$, gives $\beta_{\rm eq}$ of the document (with $\Lambda_4 = \Lambda + \kappa U_0 + \kappa f_W/V_\perp$).

The delta function and smearing. On each circle the integral $\int_{-L_c/2}^{L_c/2} e^{-2\pi i n y/L_c}\delta(y)dy = 1$ gives the Fourier series $\delta(y) = L_c^{-1}\sum_n e^{2\pi i n y/L_c}$; the zero mode of $\delta^4(\vec{y})$ is L_c^{-4}, and $\frac{1}{V_\perp}\int_F b^4\frac{\rho_4\delta^4}{b^4} = \frac{\rho_4}{V_\perp}$.

6. Verification

6.1 The checks of section 25 of 7+1.nb

Evaluation of 7+1.nb: Wolfram 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026), a fresh kernel (front-end kernel configuration Eval71); finished Wed 28 Oct 2026 08:57:47; 283.883 s; message cells 0; output cells 554. The checks of section 25 (checks 1–14 concern the torus alone, 15–16 the torus and the brane together, 17–27 the world brane):

#	check	result	section
1	auxiliary torus: for the cosmological metric det(-g_kl) == b^8 (k, l = 4..7), and the proper volume of the fundamental domain [0, Lc)^4, Integrate[Sqrt[det(-g_kl)], {y, 0, Lc}^4], is b^4 Lc^4 = V_perp	True	25.1
2	only the product b Lc of the auxiliary torus is physical: y = lambda y' maps the torus of period Lc and size b to the one of period Lc/lambda and size lambda b; the line element b^2 dy.dy, the proper period b Lc and V_perp = b^4 Lc^4 are unchanged	True	25.1
3	symmetries of the auxiliary torus that fix y = 0 (and the brane): the 384 = 2^4 4! signed permutation matrices are distinct orthogonal integer matrices (they map Z^4 onto itself and preserve |n|^2); the shells |n|^2 = 1, 2, 3, 4, 9 split into orbits of sizes {8}, {24}, {32}, {8, 16}, {8, 96}	True	25.1
4	Fourier modes of the auxiliary torus: phi_n = Exp[I q_n.y]/Sqrt[V_perp] with q_n = 2 Pi n/Lc (n in Z^4) are orthonormal for the proper volume element b^4 d^4y (exact integrals: n = m = (1, 0, -2, 3); n = (1, 0, -2, 3), m = (1, 1, -2, 3); n = 0, m = (0, 2, 0, 0))	True	25.2
5	Parseval on the auxiliary torus: f(y) = Prod_k cos^2(Pi y_k/Lc) = Prod_k (1/2 + (1/4) e^(2 Pi I y_k/Lc) + (1/4) e^(-2 Pi I y_k/Lc)) has the 81 non-zero coefficients c_n = Sqrt[V_perp] Prod_k w(n_k), w(0) = 1/2, w(+-1) = 1/4 (three of them checked by integration), and Sum_n |c_n|^2 == Int b^4 d^4y f^2 == (3/8)^4 V_perp	True	25.2
6	the Laplacian of the auxiliary torus: -(1/b^2) Sum_k d^2 phi_n/dy_k^2 == (|q_n|^2/b^2) phi_n (n = (2, -1, 0, 3))	True	25.2
7	scalar Kaluza-Klein masses: for the static metric diag(1, -1, -1, -1, -b0^2, ..., -b0^2) the Klein-Gordon operator Box + m^2 (Box f = |g|^(-1/2) d_A(|g|^(1/2) g^AB d_B f)) on Exp[-I E x0 + I k.x + I q.y] gives (-E^2 + |k|^2 + |q|^2/b0^2 + m^2) times the wave: in ordinary spacetime a particle of mass Sqrt[m^2 + |q|^2/b0^2]	True	25.2
8	Jacobi's four-square theorem: the number of n in Z^4 with |n|^2 = k is r_4(k) = 8 sigma(k) - 32 sigma(k/4) for k = 1..100 (direct count over {-10, ..., 10}^4)	True	25.3
9	Weyl's law: the number N(R) of n in Z^4 with |n| <= R approaches the volume Pi^2 R^4/2 of the four-ball: N(10) = 49689, N(100) = 493490641, and |N(R)/(Pi^2 R^4/2) - 1| < 10^-2, 10^-3, 3 10^-5, 5 10^-6, 10^-6 for R = 10, 30, 100, 300, 1000	True	25.3
10	the auxiliary torus today (M8 = 10 TeV, M_Pl = 2.435 10^18 GeV): V_perp = M_Pl^2/M8^6 = 5.929 10^12 GeV^-4, l_perp = V_perp^(1/4) = b0 Lc = 1560.4 GeV^-1 = 3.079 10^-13 m, lowest Kaluza-Klein mass 2 Pi/l_perp = 4.027 MeV; N(mu) = V_perp mu^4/(32 Pi^2) = 1.17 10^13 levels below m_r/2 = 5 GeV (the lattice count of section 22.2 agrees to 10^-5) and N(M8) = M_Pl^2/(32 Pi^2 M8^2)	True	25.3
11	Newton's law in eight dimensions: Omega_6 = 2 Pi^(7/2)/Gamma(7/2) = 16 Pi^3/15, r^-5 is harmonic away from r = 0 in seven space dimensions ((1/r^6) d/dr (r^6 d/dr r^-5) = 0), and grad^2 Phi = 8 Pi G8 ((D - 3)/(D - 2)) M delta^7 (D = 8) gives Phi = -(5/6) kappa M/(5 Omega_6 r^5) = -5 kappa M/(32 Pi^3 r^5); the same formula in D = 4 gives -G M/r	True	25.4

#	check	result	section
12	images = Kaluza-Klein modes (Poisson summation; lengths in units of l_perp): Sum_n (r^2 + |n|^2)^(-5/2) == (4 Pi^2/(3 r)) Sum_m Exp[-2 Pi r |m|] to 10^-8 at r = 1/4, 1/2, 1, 2 (left: shells with r_4(k) to k = 1.5 10^6 and the continuum tail Pi^2 Int k (r^2 + k)^(-5/2) dk; right: shells to k = 20000)	True	25.4
13	near and far field: Z(5/2) = Sum_(n != 0) |n|^-5 = Sum_k r_4(k) k^(-5/2) = 7 zeta(5/2) zeta(3/2) = 24.531 (shell sum with tail, to 10^-8); near the source r^5 S(r) - 1 -> r^5 Z(5/2) (the ratio lies between 0.98 and 1 at r = 1/10 and 1/20, closer to 1 at 1/20): the eight-dimensional law; far from it 3 r S/(4 Pi^2) - 1 -> 8 Exp[-2 Pi r], the 8 lightest Kaluza-Klein modes (ratio 1 to 2 10^-2 at r = 2 and 3)	True	25.4
14	graviton + radion: far from the source the images add up to (5 kappa M/(32 Pi^3)) (4 Pi^2/(3 V_perp r)) == (5/3) G4 M/r with G4 = kappa/(8 Pi V_perp); the radion of section 20.3, coupled to matter by (phi/(Sqrt[3] M_Pl)) T, gives the potential -(1/(Sqrt[3] M_Pl))^2 M M' Exp[-m_r r]/(4 Pi r), and (1/(Sqrt[3] M_Pl))^2/(4 Pi) == (2/3) G4 (8 Pi G4 = 1/M_Pl^2): graviton 1 + massless radion 2/3 = 5/3	True	25.4
15	Kaluza-Klein momentum is conserved in the bulk and not at the brane: Int b^4 d^4y phi_n phi_m phi_l^* == 1/Sqrt[V_perp] for l = n + m and 0 otherwise (exact: n = (1, 0, 0, 2), m = (0, 1, -1, 0), l = (1, 1, -1, 2) and (1, 1, -1, 1)); at y = 0, phi_n phi_m phi_l^* == V_perp^(-3/2) for every n, m, l	True	25.5
16	the delta function of the torus: Int_(-Lc/2)^(Lc/2) Exp[-I q_n y] delta(y) dy == 1 for every integer n, so delta^4(y) = Lc^-4 Sum_n Exp[I q_n.y] (every mode with the weight 1/Lc^4); a brane source rho4 delta^4(y)/b^4 has Int b^4 d^4y (rho4 delta^4(y)/b^4) == rho4, i.e. the average rho4/V_perp over the torus	True	25.5
17	the world brane W = {x4 = x5 = x6 = x7 = 0}: the induced metric h = J^T.g.J of the embedding X = (x0, x1, x2, x3, 0, 0, 0, 0) is, for the cosmological metric of section 11.2, the Friedmann-Robertson-Walker metric diag(1, -a^2, -a^2, -a^2) (independent of b); for the inhomogeneous metric of section 11.3 it is diag(n^2, -a^2, -a^2, -a^2) at x4 = 0 with n = 1 + x0^2/11, a = 1 + x0/3	True	25.6
18	brane integrals in eight-dimensional form: at the brane Sqrt|det g| == Sqrt|det h| b^4 with b^4 = Sqrt[det(-g_kl)] (cosmological metric; inhomogeneous metric exactly at x0 = 1/2), so Int_W d^4x Sqrt|h| L == Int d^8X Sqrt|g| (delta^4(y)/b^4) L; test: Int d^4y b^4 (delta^4(y)/b^4) F(y) == F(0) = 1 for F = Exp[x4 + 2 x5] Cos[x6 - x7]	True	25.6
19	how the brane sits in the bulk: the Christoffel symbols {k mu nu} at the brane (k = 4..7, mu, nu = 0..3; = (1/(2 b^2)) d_k g_mu nu for these metrics, b times the extrinsic curvature) all vanish for the cosmological metric: the brane is totally geodesic, and a free particle moving along it stays on it. For the inhomogeneous metric at x0 = 1/2 the only non-zero one is {4 00} = n (d_4 n)/b^2 = 1458/58619: a free particle at rest on the brane is accelerated off it	True	25.6
20	explicit forms: every Gamma17[A] is a signed permutation matrix (one entry +-1 in each row and column), and 2 Sqrt[2] U has only the entries 0, +-1, +-I, eight non-zero ones in each row and in each column	True	25.7
21	brane spinor (x) transverse spinor: Gamma17[mu].Gamma17[nu] == U.(gamma_mu.gamma_nu (x) 1_4).U^dag, Gamma17[k].Gamma17[l] == U.(1_4 (x) rho_k.rho_l).U^dag and Gamma17[mu].Gamma17[k] == U.(gamma_mu.gamma5 (x) rho_k).U^dag (mu, nu = 0..3, k, l = 4..7): the Lorentz group of the brane acts on the brane spinor index, the rotations about the brane on the transverse index, and the generators that mix them are broken by the brane	True	25.7
22	the transverse index carries Spin(4) = SU(2) x SU(2): rho4567 == -gamma5 (4 x 4), it commutes with every rho_k.rho_l and anticommutes with every rho_k; for each even permutation (k, l, m, n) of (4, 5, 6, 7) the combination rho_k.rho_l + s rho_m.rho_n vanishes on the eigenspace rho4567 = s, and rho_k.rho_l - s rho_m.rho_n acts there with rank 2: the four values of a form (2, 1) + (1, 2)	True	25.7
23	chirality at the brane: PL17 == U.(P_L (x) P_+ + P_R (x) P_-).U^dag with P_L,R = (1 -+ gamma5)/2 and P_+- = (1 +- rho4567)/2: the left-handed half psi_L (chi17 = -1, 8 components) is two brane-left-handed Weyl spinors with rho4567 = +1 and two brane-right-handed ones with rho4567 = -1 (each part of rank 4)	True	25.7
24	an explicit spinor: psi = U.(u (x) e_1) with the brane spinor u = (1, 2, 0, -1): psibar psi == ubar u == 4 and psibar Gamma^mu psi == ubar gamma^mu u (mu = 0..3)	True	25.7
25	brane tension: for S_W = -Int d^4x Sqrt|h| f_W and h = diag(1/u0, -1/u1, -1/u2, -1/u3) (h^00 = u0, h^ii = -u_i), T_mu nu = (2/Sqrt|h|) dS_W/dh^mu nu == f_W h_mu nu, T^0_0 == f_W (energy density) and T^1_1 == f_W (pressure -f_W: w = -1 along the brane)	True	25.8
26	sources smeared over the torus (W(a, b) the comoving energy, section 20.2): the brane tension W = a^3 f_W/Lc^4 gives eps = f_W/V_perp, pa = -eps, pb = 0; brane radiation W = a^3 r0 a^-4/Lc^4 gives pa = eps/3, pb = 0; radiation of the bulk, isotropic in the seven space directions, has <cos^2 theta> = 3/7 and <sin^2 theta> = 4/7 for the weight cos^2 theta sin^3 theta of the momentum sphere, so pa = (1/3)(3/7) eps = eps/7 and pb = (1/4)(4/7) eps = eps/7	True	25.8

#	check	result	section
27	the brane in four-dimensional physics: (i) the tension is vacuum energy, kappa f_W/ V_perp == f_W/M_Pl^2 (V_perp = kappa M_Pl^2), and f_W = M8^4 is between 3 10^62 and 5 10^62 times rho_Lambda = 3 H0^2 M_Pl^2 Omega_Lambda (section 24.5); (ii) normalcy temperature: (T^7/M8^6)/(T^2/M_Pl) == 1 at T* = (M8^6/M_Pl)^(1/5) = 13.3 GeV, and V_ perp T^4 T^3/M_Pl^2 == T^7/M8^6; (iii) thin brane: (1/y^3) d/dy (y^3 d/dy y^-2) == 0 and 2 Pi^2 y^3 d/dy y^-2 == -4 Pi^2 (the Green's function -1/(4 Pi^2 |y|^2) of the four transverse directions), and Sqrt[kappa f_W] == 1/M8 for f_W = M8^4	True	25.8

The comparison with the 4-sphere (section 10 of the document) rests on section 26 of 7+1.nb: 33 checks, 33 True; the independent program tools/ref_radion.py repeats them with 31 boolean checks, 31 True, and the 15 groups of numbers printed by both agree to a largest relative difference of 8.2×10^{-11}.

6.2 The independent Python program

tools/ref_torus_brane.py — its boolean checks:

#	check	result
1	torus: det(-g_kl) = b^8 for the cosmological metric; the proper volume of [0, Lc)^4 is b^4 Lc^4	True
2	only b Lc is physical: y = lambda y' maps (Lc, b) to (Lc/lambda, lambda b); b^2 dy.dy, b Lc and b^4 Lc^4 unchanged	True
3	symmetries fixing y = 0: 384 = 2^4 4! distinct signed permutation matrices, orthogonal integer matrices; shells 1, 2, 3, 4, 9 have orbits {8}, {24}, {32}, {8, 16}, {8, 96}	True
4	Fourier modes phi_n = exp(i q_n.y)/sqrt(V_perp), q_n = 2 pi n/Lc: orthonormal for b^4 d^4y (three exact pairs)	True
5	Parseval: f = prod cos^2(pi y_k/Lc) has 81 coefficients sqrt(V_perp) prod w(n_k) (3 checked by integration), Sum |c_n|^2 = Int b^4 f^2 = (3/8)^4 V_perp	True
6	Laplacian of the torus: -(1/b^2) Sum d^2/dy_k^2 phi_n = (|q_n|^2/b^2) phi_n	True
7	scalar Kaluza-Klein masses: (Box + m^2) exp(-iEx0 + ik.x + iq.y) = (-E^2 + |k|^2 + |q|^2/b0^2 + m^2) exp(...) on diag(1,-1,-1,-1,-b0^2,...)	True
8	graviton on the torus: D(D-3)/2 = 20 (D = 8) = 2 + 8 + 10 (zero mode: graviton, 4 graviphotons, 10 moduli) = 5 + 9 + 6 (massive level)	True
9	Jacobi: #{n in Z^4 : |n|^2 = k} = r_4(k) = 8 sigma(k) - 32 sigma(k/4), direct count for k = 1..100	True
10	Weyl's law: N(10) = 49689, N(100) = 493490641; |N(R)/(pi^2 R^4/2) - 1| < 1e-2, 1e-3, 3e-5, 5e-6, 1e-6	True
11	numbers today: V_perp = 5.929e12 GeV^-4, l_perp = 1560.4 GeV^-1 = 3.079e-13 m, 2 pi/l_perp = 4.027 MeV; N(5 GeV) = 1.17e13 (lattice to 1e-5); N(M8) = M_Pl^2/(32 pi^2 M8^2)	True
12	from eight to four dimensions: for diag(1, -a^2, -a^2, -a^2, -b0^2, ..., -b0^2) (static torus) R = R4 = -6(Ha' + 2 Ha^2), the scalar curvature of the FRW metric diag(1, -a^2, -a^2, -a^2); sqrt|g| = a^3 b0^4	True
13	quaternion subalgebra of the brane directions: e1 e2 = -e3, e2 e3 = -e1, e3 e1 = -e2 (e_i e_j = -delta_ij - eps_ijk e_k), span{e0, ..., e3} closed under the octonion product, Re(e_A e_B) = diag(1, -1, -1, -1)	True
14	Newton in 8D: Omega_6 = 16 pi^3/15, (1/r^6) d/dr (r^6 d/dr r^-5) = 0, Phi = -(5/6) kappa M/(5 Omega_6 r^5) = -5 kappa M/(32 pi^3 r^5); D = 4: -G M/r	True
15	4D Fourier transform: Int d^4y exp(-ip.y) (r^2 + |y|^2)^(-5/2) = (4 pi^2/(3 r)) exp(-p r) (mpmath, three (p, r), to 1e-20)	True
16	Poisson duality: Sum_n (r^2 + |n|^2)^(-5/2) = (4 pi^2/(3 r)) Sum_m exp(-2 pi r |m|) to 1e-8 at r = 1/4, 1/2, 1, 2	True
17	Z(5/2) = 7 zeta(5/2) zeta(3/2) = Sum r_4(k) k^-5/2 (to 1e-8); near field ratio -> 1 (0.98 < . < 1 at r = 1/10, 1/20); far field ratio -> 1 (to 2e-2 at r = 2, 3)	True
18	graviton + radion: (5 kappa M/(32 pi^3))(4 pi^2/(3 V_perp r)) = (5/3) G4 M/r; radion (1/ (sqrt3 M_Pl))^2/(4 pi) = (2/3)/(8 pi M_Pl^2)	True
19	Kaluza-Klein momentum: bulk vertex Int b^4 phi_n phi_m phi_l^* = 1/sqrt(V_perp) for l = n + m, 0 otherwise; at y = 0 phi_n phi_m phi_l^* = V_perp^(-3/2)	True
20	delta function of the torus: Fourier coefficients Int exp(-i q_n y) delta(y) dy = 1; Int b^4 d^4y rho4 delta^4(y)/b^4 = rho4 (average rho4/V_perp)	True
21	induced metric h = J^T g J: FRW diag(1, -a^2, -a^2, -a^2) for the cosmological metric; diag(n^2, -a^2, -a^2, -a^2) at x4 = 0 with n = 1 + x0^2/11, a = 1 + x0/3 for the inhomogeneous metric	True
22	brane volume element: sqrt|g| = sqrt|h| b^4 at the brane (cosmological metric; inhomogeneous metric at x0 = 1/2, exact), so Int_W d^4x sqrt|h| L = Int d^8X sqrt|g| delta^4(y)/b^4 L; test Int b^4 (delta^4/b^4) F = F(0)	True

#	check	result
23	Christoffel symbols {k mu nu} at the brane: all 0 for the cosmological metric (totally geodesic); for the inhomogeneous metric at x0 = 1/2 only {4 00} = 1458/58619 = n d4n/b^2	True
24	R_A = (gamma_mu (x) 1_4, gamma5 (x) rho_k) and Gamma17[A] both satisfy X_A X_B + X_B X_A = 2 eta_AB 1_16	True
25	every Gamma17[A] is a signed permutation matrix	True
26	intertwiner: Schur average with the seed matrix unit (1, 2); 2 sqrt(2) U has entries 0, +-1, +-i (rounding error < 1e-12) with 8 non-zero entries in each row and column; Gamma17[A].U8 = U8.R_A and U8^dag.U8 = 8 1_16 exactly	True
27	Schur's lemma: the linear map X -> (Gamma_A X - X R_A)_A has a one-dimensional null space (one singular value < 1e-10, the next > 0.1)	True
28	brane Lorentz group on the brane spinor index, transverse rotations on a: Gamma_mu Gamma_nu = U (gamma_mu gamma_nu (x) 1) U^dag, Gamma_k Gamma_l = U (1 (x) rho_k rho_l) U^dag, Gamma_mu Gamma_k = U (gamma_mu gamma5 (x) rho_k) U^dag	True
29	Spin(4) on the transverse index: rho4567 = -gamma5 (4x4) commutes with rho_k rho_l, anticommutes with rho_k; (rho_k rho_l + s rho_m rho_n) P_s = 0 and (rho_k rho_l - s rho_m rho_n) P_s has rank 2 for even (k,l,m,n): 4 = (2,1) + (1,2)	True
30	chirality: chi = i Gamma_0...Gamma_7 = U (gamma5 (x) rho4567) U^dag; P_L17 = U (P_L (x) P_+ + P_R (x) P_-) U^dag, each part of rank 4	True
31	example psi = U.(u (x) e_1), u = (1, 2, 0, -1): psibar psi = ubar u = 4 and psibar Gamma^mu psi = ubar gamma^mu u	True
32	four brane Dirac spinors: psibar psi' = Sum_a psibar_a psi'_a and psibar Gamma^mu psi' = Sum_a psibar_a gamma^mu psi'_a (random integer spinors)	True
33	Kaluza-Klein operator X = gamma5 (x) (rho.q)/b0: X^2 = -|q|^2/b0^2 1_16 and X anticommutes with gamma_mu (x) 1_4	True
34	brane tension: T_mu nu = (2/sqrt|h|) dS/dh^mu nu = f_W h_mu nu for S = -Int sqrt|h| f_W; T^0_0 = f_W, T^1_1 = f_W (p = -f_W, w = -1)	True
35	smeared sources: tension eps = f_W/V_perp, pa = -eps, pb = 0; brane radiation pa = eps/3, pb = 0; bulk radiation <cos^2> = 3/7, <sin^2> = 4/7: pa = pb = eps/7	True
36	normalcy temperature: (T^7/M8^6)/(T^2/M_Pl) = 1 at T* = (M8^6/M_Pl)^(1/5) = 13.3 GeV; V_perp T^4 T^3/M_Pl^2 = T^7/M8^6	True
37	the tension is vacuum energy: kappa f_W/V_perp = f_W/M_Pl^2 (V_perp = kappa M_Pl^2); f_W = M8^4 is 3e62 to 5e62 rho_Lambda	True
38	Green's function of the four transverse directions: (1/y^3) d/dy (y^3 d/dy y^-2) = 0, flux 2 pi^2 y^3 d/dy y^-2 = -4 pi^2; sqrt(kappa f_W) = 1/M8 for f_W = M8^4	True

Numbers printed by both programs (largest relative difference per group, and the tolerance used by tools/verify_7p1_nb.py; integers and exact matrices must agree exactly):

quantity	values	max. relative difference	tolerance
orbit sizes of the shells $k = 1, \dots, 12$	16	0	exact
$r_4(k)$, $k = 0, \dots, 16$	17	0	exact
Weyl: $N(R)$, $R = 10, 30, 100, 300, 1000$	5	0	exact
Weyl: $N(R)/(\pi^2 R^4/2) - 1$	5	3.8×10^{-10}	1×10^{-9}
$V_\perp$, $\ell_\perp$ (GeV^{-1}, m, fm), $2\pi/\ell_\perp$	5	2.2×10^{-16}	1×10^{-12}
$\ell_\perp/r_p$, $a_0/\ell_\perp$, $R = \ell_\perp/2\pi$, M_D	5	1.7×10^{-16}	1×10^{-12}
$N(\mu)$ at 5 GeV, m_h, M_8; lattice count below 5 GeV	4	1.8×10^{-16}	1×10^{-12}
the first levels (k, r_4, $4r_4$, mass)	32	2.5×10^{-16}	1×10^{-12}
image sums $S(r)$ and Kaluza–Klein sums, $r = \frac{1}{4}, \frac{1}{2}, 1, 2$	8	1.2×10^{-15}	1×10^{-9}
$Z(5/2)$: closed form, shell sum	2	2.9×10^{-16}	1×10^{-9}
near-field and far-field ratios	5	7.8×10^{-12}	1×10^{-6}
Christoffel symbol $\{4\,00\}$ at the brane	1	0	1×10^{-15}
Γ_A as signed permutations	128	0	exact
$\mathrm{Re}(2\sqrt{2}\,U)$	256	0	exact
$\mathrm{Im}(2\sqrt{2}\,U)$	256	0	exact
example spinor: $\bar\psi\psi$, $\bar\psi\Gamma^\mu\psi$	5	0	1×10^{-15}
T_*, f_W/ρ_Λ, $1/(2\pi M_8)$	3	5.8×10^{-16}	1×10^{-12}

6.3 The earlier sections are unchanged

tools/verify_7p1_nb.py compares the evaluated notebook with the backup of section 3 and with the Python program; its lines on section 25:

```
section 25: check outputs (all True) : 27 (True)
stage 6: without section 25 (and with the old title of the summary section) the 786 input and
    ...text cells equal those of the backup taken before section 25 was appended : True
stage 6: the 481 outputs of sections 0-24 and of the summary equal those of the backup (summary
    ...counts and tables, the private context name and measured times excepted) : True
section 25 numbers: 17 groups of printed numbers agree with tools/ref_torus_brane.py (integers
    ...and exact values exactly, floating-point numbers to 1e-15 ... 1e-6 per group) : True
VERDICT: ALL CROSS-CHECKS PASSED
```

6.4 Files

SHA-256 of the files (at the time this document was generated):

```
7+1.nb                          41d252f9852f210c228f45e1ac956d2d7ef26faeb0062a5880280e1aae0ef
    ...fc7
tools/make_7p1_nb.py            996d26c9552cd44902560af16a1683fa71457261f18fefaeddae87bd99a2b
    ...144
tools/nb_part_gravity.py        c4caf2ad5c932e8f603341edade5f4b98fa784feb05906c051eab7c884cd6
    ...8f8
tools/nb_part_ks_gravity.py     a656c0ad040b4576b75b17e029545109e531c8e656ad99bfdb5cba946e967
    ...772
tools/nb_part_sm.py             43e92beed57453a4b57931de6d603ac6c218efba6e4005133b69b8054a975
    ...c5b
tools/nb_part_coupled.py        e630521d64132e71afc861114f699eb06454557cf14a57ad8ec1eef96247a
    ...ae0
tools/nb_part_dark.py           10ad8fb8ae25fa07bbee758c653995acd4e2c80733059a8b3648bd1593e46
    ...db0
tools/nb_part_torus_brane.py    fb904ff8d85b904c55b3b2a8f60fb0cb047f1ccf1b98db3364092f2bae0b2
    ...162
tools/nb_part_radion.py         53c5604707a3e22228c840ee0a2d2125616f9410f75a5535041dfaf18f497
    ...c00
tools/lovelock71.py             d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af9
    ...5a1
tools/ref_gravity.py            f07cf8adaff585af89c87d19532f1aaa7db606ad88ae90483b2d1f41341ed
    ...f36
tools/ref_ks_gravity.py         b6488ef24b1c16aacd8ff6cf6a5a1ba3f316e082449ad97480022d1a26383
    ...27f
tools/ref_sm.py                 f3f80806893f42bbf166e1f1de896ffd611dbfe847685da8e3ee367e9cde6
    ...aba
tools/ref_coupled.py            695dbd9375753653472d9c37502f0067c4a64fe01a64ba8c05e8b2c5da85a
    ...a58
tools/ref_dark.py               1600e6b57ab0383b4f2b664e2796d3864c9afa77a9ed59f007f66e1139453
    ...ff5
tools/ref_torus_brane.py        1f623656ab0f5c8ee43cb310f7bfd580cbba5a7385da2963adaae4d2f8bda
    ...af7
tools/ref_radion.py             e9d47176b9f2d2232a84b3b82188bd3b0a19cde3dbe1ba013c1fe5abff634
    ...1ce
tools/ref71.py                  1a7a06eb0add3f459611776dc9c474a5bb959d947a9db949db4612b1ab943
    ...4e0
tools/ref44.py                  ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064f
    ...af5
```

Output of tools/ref_torus_brane.py (independent checks of section 25; exit code 0) (complete):

```
torus: det(-g_kl) = b^8 for the cosmological metric; the proper volume of [0, Lc)^4 is b^4 Lc^4
    ...: True
only b Lc is physical: y = lambda y' maps (Lc, b) to (Lc/lambda, lambda b); b^2 dy.dy, b Lc and
    ...b^4 Lc^4 unchanged : True
symmetries fixing y = 0: 384 = 2^4 4! distinct signed permutation matrices, orthogonal integer
    ...matrices; shells 1, 2, 3, 4, 9 have orbits {8}, {24}, {32}, {8, 16}, {8, 96} : True
NUM orbit sizes of the shells k = 1..12 : [[8], [24], [32], [8, 16], [48], [96], [64], [24],
    ...[8, 96], [48, 96], [96], [32, 64]]
Fourier modes phi_n = exp(i q_n.y)/sqrt(V_perp), q_n = 2 pi n/Lc: orthonormal for b^4 d^4y
    ...(three exact pairs) : True
Parseval: f = prod cos^2(pi y_k/Lc) has 81 coefficients sqrt(V_perp) prod w(n_k) (3 checked by
    ...integration), Sum |c_n|^2 = Int b^4 f^2 = (3/8)^4 V_perp : True
Laplacian of the torus: -(1/b^2) Sum d^2/dy_k^2 phi_n = (|q_n|^2/b^2) phi_n : True
scalar Kaluza-Klein masses: (Box + m^2) exp(-iEx0 + ik.x + iq.y) = (-E^2 + |k|^2 + |q|^2/b0^2 +
    ...m^2) exp(...) on diag(1,-1,-1,-1,-b0^2,...) : True
graviton on the torus: D(D-3)/2 = 20 (D = 8) = 2 + 8 + 10 (zero mode: graviton, 4 graviphotons,
    ...10 moduli) = 5 + 9 + 6 (massive level) : True
Jacobi: #{n in Z^4 : |n|^2 = k} = r_4(k) = 8 sigma(k) - 32 sigma(k/4), direct count for k =
    ...1..100 : True
NUM r_4(k), k = 0..16 : [1, 8, 24, 32, 24, 48, 96, 64, 24, 104, 144, 96, 96, 112, 192, 192, 24]
NUM N(R) for R = 10, 30, 100, 300, 1000 : [49689, 4000425, 493490641, 39972070929,
    ...4934805110729]
NUM N(R)/(pi^2 R^4/2) - 1 for R = 10, 30, 100, 300, 1000 : [0.006909658800824392,
    ...0.0008093730182694436, 2.1117250719626668e-05, 4.330657224604906e-06,
    ...5.897266401877488e-07]
Weyl's law: N(10) = 49689, N(100) = 493490641; |N(R)/(pi^2 R^4/2) - 1| < 1e-2, 1e-3, 3e-5,
    ...5e-6, 1e-6 : True
NUM V_perp (GeV^-4), l_perp (GeV^-1), l_perp (m), l_perp (fm), 2 pi/l_perp (MeV) :
    ...(5929225000000.0, 1560.4486534327234, 3.079186208511254e-13, 307.9186208511254,
    ...4.026524867292262)
NUM l_perp/proton radius, Bohr radius/l_perp, R = l_perp/(2 pi) (GeV^-1, m), M_D = (2 pi)^(2/3)
    ...M8 (GeV) : (365.9598536381334, 171.8561902623129, 248.3531166349098,
    ...4.9006770578495764e-14, 34050.219214767545)
NUM N(mu) = V_perp mu^4/(32 pi^2) for mu = 5 GeV, m_h, M8; lattice count below 5 GeV :
    ...(11733517482065.238, 4.61280953636634e+18, 1.877362797130438e+26, 11733511217161)
numbers today: V_perp = 5.929e12 GeV^-4, l_perp = 1560.4 GeV^-1 = 3.079e-13 m, 2 pi/l_perp =
    ...4.027 MeV; N(5 GeV) = 1.17e13 (lattice to 1e-5); N(M8) = M_Pl^2/(32 pi^2 M8^2) : True
NUM first levels: k, r_4(k), Dirac fermions 4 r_4(k), mass 2 pi sqrt(k)/l_perp (MeV), k = 1..8
    ...: [(1, 8, 32, 4.026524867292262), (2, 24, 96, 5.694366076557245), (3, 32, 128,
    ...6.9741456480897295), (4, 24, 96, 8.053049734584524), (5, 48, 192, 9.003583316358819),
    ...(6, 96, 384, 9.862931361493793), (7, 64, 256, 10.65318344667268), (8, 24, 96,
    ...11.38873215311449)]
from eight to four dimensions: for diag(1, -a^2, -a^2, -a^2, -b0^2, ..., -b0^2) (static torus)
    ...R = R4 = -6(Ha' + 2 Ha^2), the scalar curvature of the FRW metric diag(1, -a^2, -a^2,
    ...-a^2); sqrt|g| = a^3 b0^4 : True
quaternion subalgebra of the brane directions: e1 e2 = -e3, e2 e3 = -e1, e3 e1 = -e2 (e_i e_j =
    ...-delta_ij - eps_ijk e_k), span{e0, ..., e3} closed under the octonion product, Re(e_A
    ...e_B) = diag(1, -1, -1, -1) : True
NUM quaternion products e1 e2, e2 e3, e3 e1: coefficients of e3, e1, e2 : (-1, -1, -1)
Newton in 8D: Omega_6 = 16 pi^3/15, (1/r^6) d/dr (r^6 d/dr r^-5) = 0, Phi = -(5/6) kappa M/(5
    ...Omega_6 r^5) = -5 kappa M/(32 pi^3 r^5); D = 4: -G M/r : True
4D Fourier transform: Int d^4y exp(-ip.y) (r^2 + |y|^2)^(-5/2) = (4 pi^2/(3 r)) exp(-p r)
    ...(mpmath, three (p, r), to 1e-20) : True
NUM images S(r), Kaluza-Klein (4 pi^2/(3 r)) Sum exp(-2 pi r |m|), relative difference, at r =
    ...1/4, 1/2, 1, 2 : [(1046.8493759696448, 1046.8493759704088, -7.297495940861154e-13),
    ...(51.12525179526186, 51.12525179602524, -1.4931500480486193e-11), (13.40960454140553,
    ...13.409604542168916, -5.692823989889938e-11), (6.579922931695463, 6.579922932458839,
    ...-1.1601597460497715e-10)]
Poisson duality: Sum_n (r^2 + |n|^2)^(-5/2) = (4 pi^2/(3 r)) Sum_m exp(-2 pi r |m|) to 1e-8 at
    ...r = 1/4, 1/2, 1, 2 : True
NUM Z(5/2) = 7 zeta(5/2) zeta(3/2), shell sum with tail : (24.53127768992603,
    ...24.531277689162653)
NUM near field (r^5 S - 1)/(r^5 Z(5/2)) at r = 1/20, 1/10; far field (3 r S/(4 pi^2) - 1)/(8
    ...exp(-2 pi r)) at r = 1, 2, 3 : [0.9970259928360821, 0.9882280058562034,
```

```
    ...1.272311659711708, 1.016880665840871, 1.0012238149051904]
Z(5/2) = 7 zeta(5/2) zeta(3/2) = Sum r_4(k) k^-5/2 (to 1e-8); near field ratio -> 1 (0.98 < . <
    ...1 at r = 1/10, 1/20); far field ratio -> 1 (to 2e-2 at r = 2, 3) : True
graviton + radion: (5 kappa M/(32 pi^3))(4 pi^2/(3 V_perp r)) = (5/3) G4 M/r; radion (1/(sqrt3
    ...M_Pl))^2/(4 pi) = (2/3)/(8 pi M_Pl^2) : True
Kaluza-Klein momentum: bulk vertex Int b^4 phi_n phi_m phi_l^* = 1/sqrt(V_perp) for l = n + m,
    ...0 otherwise; at y = 0 phi_n phi_m phi_l^* = V_perp^(-3/2) : True
delta function of the torus: Fourier coefficients Int exp(-i q_n y) delta(y) dy = 1; Int b^4
    ...d^4y rho4 delta^4(y)/b^4 = rho4 (average rho4/V_perp) : True
induced metric h = J^T g J: FRW diag(1, -a^2, -a^2, -a^2) for the cosmological metric;
    ...diag(n^2, -a^2, -a^2, -a^2) at x4 = 0 with n = 1 + x0^2/11, a = 1 + x0/3 for the
    ...inhomogeneous metric : True
brane volume element: sqrt|g| = sqrt|h| b^4 at the brane (cosmological metric; inhomogeneous
    ...metric at x0 = 1/2, exact), so Int_W d^4x sqrt|h| L = Int d^8X sqrt|g| delta^4(y)/b^4 L;
    ...test Int b^4 (delta^4/b^4) F = F(0) : True
Christoffel symbols {k mu nu} at the brane: all 0 for the cosmological metric (totally
    ...geodesic); for the inhomogeneous metric at x0 = 1/2 only {4 00} = 1458/58619 = n d4n/b^2
    ...: True
NUM Christoffel {4 00} at the brane (inhomogeneous metric, x0 = 1/2) : 0.02487248161858783
R_A = (gamma_mu (x) 1_4, gamma5 (x) rho_k) and Gamma17[A] both satisfy X_A X_B + X_B X_A = 2
    ...eta_AB 1_16 : True
every Gamma17[A] is a signed permutation matrix : True
NUM Gamma17[A] as signed permutations: column j has the entry sign(c) in row |c|, c listed for
    ...A = 0..7 : [[9, 10, 11, 12, 13, 14, 15, 16, 1, 2, 3, 4, 5, 6, 7, 8], [-8, -7, 6, 5, -4,
    ...-3, 2, 1, 16, 15, -14, -13, 12, 11, -10, -9], [7, -8, -5, 6, 3, -4, -1, 2, -15, 16, 13,
    ...-14, -11, 12, 9, -10], [-6, 5, -8, 7, -2, 1, -4, 3, 14, -13, 16, -15, 10, -9, 12, -11],
    ...[-2, 1, 4, -3, 6, -5, -8, 7, 10, -9, -12, 11, -14, 13, 16, -15], [3, 4, -1, -2, -7, -8,
    ...5, 6, -11, -12, 9, 10, 15, 16, -13, -14], [4, -3, 2, -1, -8, 7, -6, 5, -12, 11, -10, 9,
    ...16, -15, 14, -13], [5, 6, 7, 8, -1, -2, -3, -4, -13, -14, -15, -16, 9, 10, 11, 12]]
intertwiner: Schur average with the seed matrix unit (1, 2); 2 sqrt(2) U has entries 0, +-1,
    ...+-i (rounding error < 1e-12) with 8 non-zero entries in each row and column;
    ...Gamma17[A].U8 = U8.R_A and U8^dag.U8 = 8 1_16 exactly : True
Schur's lemma: the linear map X -> (Gamma_A X - X R_A)_A has a one-dimensional null space (one
    ...singular value < 1e-10, the next > 0.1) : True
NUM Re(2 sqrt(2) U) : [[0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0], [0, 0, 0, 1, -1, 0, 0,
    ...0, 0, 0, 0, -1, 1, 0, 0, 0], [0, 0, 1, 0, 0, -1, 0, 0, 0, 0, 1, 0, 0, -1, 0, 0], [1, 0,
    ...0, 0, 0, 0, 0, 1, -1, 0, 0, 0, 0, 0, 0, -1], [0, 0, 0, -1, -1, 0, 0, 0, 0, 0, 0, -1, -1,
    ...0, 0, 0], [0, -1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, -1, 0], [-1, 0, 0, 0, 0, 0, 0, 1,
    ...-1, 0, 0, 0, 0, 0, 0, 1], [0, 0, -1, 0, 0, -1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0], [0, 1, 0,
    ...0, 0, 0, 1, 0, 0, -1, 0, 0, 0, 0, -1, 0], [0, 0, 0, 1, -1, 0, 0, 0, 0, 0, 0, 1, -1, 0,
    ...0, 0], [0, 0, 1, 0, 0, -1, 0, 0, 0, 0, -1, 0, 0, 1, 0, 0], [1, 0, 0, 0, 0, 0, 0, 1, 1,
    ...0, 0, 0, 0, 0, 0, 1], [0, 0, 0, -1, -1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0], [0, -1, 0, 0,
    ...0, 0, 1, 0, 0, -1, 0, 0, 0, 0, 1, 0], [-1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0,
    ...-1], [0, 0, -1, 0, 0, -1, 0, 0, 0, 0, -1, 0, 0, -1, 0, 0]]
NUM Im(2 sqrt(2) U) : [[0, 0, 0, -1, -1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0], [0, 1, 0, 0, 0, 0,
    ...-1, 0, 0, 1, 0, 0, 0, 0, -1, 0], [-1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, -1], [0,
    ...0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0], [0, 1, 0, 0, 0, 0, 1, 0, 0, -1, 0, 0, 0,
    ...0, -1, 0], [0, 0, 0, -1, 1, 0, 0, 0, 0, 0, 0, -1, 1, 0, 0, 0], [0, 0, 1, 0, 0, -1, 0, 0,
    ...0, 0, -1, 0, 0, 1, 0, 0], [-1, 0, 0, 0, 0, 0, 0, -1, -1, 0, 0, 0, 0, 0, 0, -1], [0, 0,
    ...0, -1, -1, 0, 0, 0, 0, 0, 0, -1, -1, 0, 0, 0], [0, 1, 0, 0, 0, 0, -1, 0, 0, -1, 0, 0, 0,
    ...0, 1, 0], [-1, 0, 0, 0, 0, 0, 0, 1, -1, 0, 0, 0, 0, 0, 0, 1], [0, 0, 1, 0, 0, 1, 0, 0,
    ...0, 0, -1, 0, 0, -1, 0, 0], [0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0], [0, 0, 0,
    ...-1, 1, 0, 0, 0, 0, 0, 0, 1, -1, 0, 0, 0], [0, 0, 1, 0, 0, -1, 0, 0, 0, 0, 1, 0, 0, -1,
    ...0, 0], [-1, 0, 0, 0, 0, 0, 0, -1, 1, 0, 0, 0, 0, 0, 0, 1]]
brane Lorentz group on the brane spinor index, transverse rotations on a: Gamma_mu Gamma_nu = U
    ...(gamma_mu gamma_nu (x) 1) U^dag, Gamma_k Gamma_l = U (1 (x) rho_k rho_l) U^dag, Gamma_mu
    ...Gamma_k = U (gamma_mu gamma5 (x) rho_k) U^dag : True
Spin(4) on the transverse index: rho4567 = -gamma5 (4x4) commutes with rho_k rho_l,
    ...anticommutes with rho_k; (rho_k rho_l + s rho_m rho_n) P_s = 0 and (rho_k rho_l - s
    ...rho_m rho_n) P_s has rank 2 for even (k,l,m,n): 4 = (2,1) + (1,2) : True
chirality: chi = i Gamma_0...Gamma_7 = U (gamma5 (x) rho4567) U^dag; P_L17 = U (P_L (x) P_+ +
    ...P_R (x) P_-) U^dag, each part of rank 4 : True
example psi = U.(u (x) e_1), u = (1, 2, 0, -1): psibar psi = ubar u = 4 and psibar Gamma^mu psi
    ...= ubar gamma^mu u : True
NUM example psi = U.(u (x) e_1), u = (1, 2, 0, -1): psibar psi, psibar Gamma^mu psi (mu = 0..3)
```

```
    ...: [4, 6.0, 2.0, 0.0, -4.0]
four brane Dirac spinors: psibar psi' = Sum_a psibar_a psi'_a and psibar Gamma^mu psi' = Sum_a
    ...psibar_a gamma^mu psi'_a (random integer spinors) : True
Kaluza-Klein operator X = gamma5 (x) (rho.q)/b0: X^2 = -|q|^2/b0^2 1_16 and X anticommutes with
    ...gamma_mu (x) 1_4 : True
brane tension: T_mu nu = (2/sqrt|h|) dS/dh^mu nu = f_W h_mu nu for S = -Int sqrt|h| f_W; T^0_0
    ...= f_W, T^1_1 = f_W (p = -f_W, w = -1) : True
smeared sources: tension eps = f_W/V_perp, pa = -eps, pb = 0; brane radiation pa = eps/3, pb =
    ...0; bulk radiation <cos^2> = 3/7, <sin^2> = 4/7: pa = pb = eps/7 : True
normalcy temperature: (T^7/M8^6)/(T^2/M_Pl) = 1 at T* = (M8^6/M_Pl)^(1/5) = 13.3 GeV; V_perp
    ...T^4 T^3/M_Pl^2 = T^7/M8^6 : True
NUM T* = (M8^6/M_Pl)^(1/5) (GeV), f_W/rho_Lambda for f_W = M8^4, 1/(2 pi M8) (GeV^-1) :
    ...(13.264784746562139, 3.9709863911436803e+62, 1.5915494309189534e-05)
the tension is vacuum energy: kappa f_W/V_perp = f_W/M_Pl^2 (V_perp = kappa M_Pl^2); f_W = M8^4
    ...is 3e62 to 5e62 rho_Lambda : True
Green's function of the four transverse directions: (1/y^3) d/dy (y^3 d/dy y^-2) = 0, flux 2
    ...pi^2 y^3 d/dy y^-2 = -4 pi^2; sqrt(kappa f_W) = 1/M8 for f_W = M8^4 : True
boolean checks: 37   failed: 0
VERDICT: ALL CHECKS PASSED
```